

Rishi Vora

Final Year BS-MS Student in Physics at IISER Mohali

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Education

2021 — 2026	Indian Institute of Science Education and Research (IISER), Mohali INTEGRATED BACHELORS & MASTERS IN PHYSICS, WITH MINOR IN DATA SCIENCE Relevant Courses: Foundations of Quantum Mechanics and Quantum Information Advanced Quantum Mechanics Machine Learning and Artificial Intelligence Introduction to Data Science Algorithms and Complexity Computational Methods in Physics Modelling Complex Systems Nonlinear Dynamics, Chaos, and Complex Systems Theory of Computation
2021	eduNova Science Higher Secondary School, Ahmedabad 12TH — GUJARAT SECONDARY AND HIGHER SECONDARY EDUCATION BOARD Grade: A2 Percentile: 92.25

Research Experience

12/2025 — Present	Generative quantum machine learning, IISER Pune MASTER THESIS Guide: Prof. Santhanam Code: https://github.com/Vortriz/thesis ▶ Created a new Generative QML model that requires fewer resources and has significantly faster convergence compared to existing models, such as QuDDPM. ▶ Applied it to various tasks, such as learning many-body phases, states distributed over nontrivial topology, localized states of Quantum Kicked Rotor etc. ▶ Benchmarked gradient-free optimization strategies such as QNSPSA, SPSA, and Rotosolve against gradient-based approaches. ▶ Created a Julia implementation using Yao.jl, with optional CUDA acceleration.
Summer 2024	Reading project on QKD protocols, IISER Mohali SUMMER READING PROJECT Guide: Prof. Arvind ▶ Reviewed working of BB84, B92 and E91 QKD protocols. ▶ Worked out Shor and Preskill's security proof of BB84 protocol.
Summer 2023	Quantum algorithms and circuit decomposition, IISER Mohali SUMMER INTERNSHIP Guide: Dr. Sandeep Goyal Code: https://github.com/Vortriz/Summer2023 ▶ Used Qiskit to implement basic circuits like Full Adder, Multiplier, Superposition Adder, and XOR adder. ▶ Attempted a circuit decomposition scheme using Solovay-Kitaev algorithm.

Skills & Certifications

2024	Qiskit Global Summer School 2024 IBM ▶ Certificate of Quantum Excellence ▶ Lectures, interactive sessions, and programming tasks on Circuit Transpilation, Modelling and Characterization of Hardware Noise, Execution on Noisy Quantum Hardware using Qiskit.
2023	Qiskit Global Summer School 2023 IBM ▶ Certificate of Quantum Excellence ▶ Lectures, interactive sessions, and programming tasks on Dynamic circuits, Phase estimation, Factoring, Error mitigation, VQE, and Iterative phase estimation.
Programming & Tools	Python (Qiskit, PyTorch, TensorCircuit) Julia (Yao.jl) Linux (NixOS) Typst Git Some of the code I have written as a part of coursework can be found on my website .
Spoken Languages	Fluent in English , Hindi , as well as Gujarati .
Hobbies	I actively develop and contribute to FOSS projects on GitHub , create websites once in a while, and spend my downtime listening to music and watching eclectic films.

Scholarships

2022 — Present	IISER Mohali — MERIT-CUM-MEANS SCHOLARSHIP Bi-annual scholarship supporting Bachelor's and Master's studies.
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Extracurriculars

June 2026	Won 2 bounties in unitaryHACK 2026 Annual virtual hackathon organized by Unitary Foundation for contributing to open-source quantum ecosystem.
March 2026	Conducted Workshops on Python Tooling, Julia, and Typst IISER PUNE ▶ Link: https://github.com/Vortriz/workshops ▶ Conducted a series of interactive workshops focused on addressing problems with Python tooling ecosystem, introducing Julia as a performant alternative for scientific computing, and showcasing Typst for high-quality document preparation.
2023 — 2024	Convenor of Turing Club (Computation Club) IISER MOHALI Organized Hackathons, Workshops, Talks etc. Took programming help sessions.